

SIMATS

ENGINEERING

SIMATS Online Programming Lab

PYTHON PROGRAMMING

SAFRIN RITHA AM

192313059RC

Q1

Prime Number

10 marks

✓

Q2

Fibonacci series

10 marks

✓

Q3

String

10 marks

✓

Q4

ODD or EVEN

10 marks

✓

Q5

Print numbers from 1 to 10

10 marks

✓

5/5 answered

A text processing system receives user input in the form of a string that may contain alphabets, digits, and special characters. Write a Python program that:

- Accepts a string input from the user
- Counts the number of vowels, consonants, digits, and special characters present in the string
- Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
1 s = input()
2 vowels = 0
3 consonants = 0
4 digits = 0
5 special = 0
6 for ch in s:
7     if ch.isalpha():
8         if ch.lower() in "aeiou":
9             vowels += 1
10        elif ch.isdigit():
11            digits += 1
12        else:
13            special += 1
14 print("vowels:", vowels)
15 print("consonants", consonants)
16 print("Digits:", digits)
17 print("special character:", special)
```

Input

hello@#

Output

```
vowels: 2
consonants 0
Digits: 0
special character: 3
```

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String

10 marks

✓

Q4

ODD or EVEN

10 marks

✓

Q5

Print numbers from 1 to 10

10 marks

✓

5/5 answered

A program is designed to check whether a number entered by the user is even or odd. However, the program contains syntax and logical errors. Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

```
num = input("Enter number:")
if num % 2 == 0
print("Even")
else
print("Odd")
```

Your Code

```
1 n = int(input())
2 if n % 2 == 0:
3     print("Even")
4 else:
5     print("odd")
```

Input

12

Output

Even

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QUESTIONS

Q1

Prime Number

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Q2

Fibonacci series

10 marks

Q3

String

10 marks

Q4

ODD or EVEN

10 marks

Q5

Print numbers from 1 to 10

10 marks

5/5 answered

Q5

Print numbers from 1 to 10

10 marks

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing. Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

```
for i in _____:
    print(_____)
```

Also explain how the range() function works in this context.

Your Code

```
1 for n in range(1,11):
2     if n > 1:
3         prime = True
4         for i in range(2,int(n**0.5) + 1):
5             if n % i == 0:
6                 prime = False
7                 break
8             if prime:
9                 print(n)
```

Input

stdin input

Output

```
2
3
5
7
```

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QUESTIONS

Q1

Harshad number

10 marks

Q2

GCD

10 marks

Q3

sum to a target value

10 marks

Q4

Bitwise Operators

10 marks

Q5

binary search

10 marks

5/5 answered

Q1

Harshad number

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, 18 % 9 == 0. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
n = int(input())
temp = n
s = 0
while temp > 0:
    s += temp % 10
    temp = temp // 10
```

Save Answer

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QUESTIONS

Q1Find smallest and largest number10 marks

Q2Missing Number10 marks

Q3Second largest number10 marks

Q4Most repeated number10 marks

Q5Unique elements10 marks

Q6Rotate an array by 'N' positions10 marks

Q7Remove all duplicates in an array10 marks

5/5 answered

Q1Find smallest and largest number10 marks

WAP to find smallest and largest number in an array

input: 12 5 18 3 25 9

output:

Smallest number:3

Largest number:25

Your Code

```
1 arr = list(map(int,input().split()))
2
3 small = min(arr)
4 large = max(arr)
5
6 print("Smallest number:", small)
7 print("Largest number:", large)
```

Input

25 3

Output

```
Smallest number: 3
Largest number: 25
```

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QUESTIONS

Q1Harshad number10 marks

Q2GCD10 marks

Q3sum to a target value10 marks

Q4Bitwise Operators10 marks

Q5binary search10 marks

5/5 answered

Q2GCD10 marks

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

```
a = int(input("Enter first: ")) b = int(input("Enter second: "))
while b = 0: temp = a a = b b = temp % a
print("GCD is", a)
```

✶ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 a = int(input("Enter first: "))
2 b = int(input("Enter second: "))
3 while b != 0:
4     temp = a
5     a = b
6     b = temp % b
7 print("GCD is", a)
```

Test Input

12
13

Output

```
Enter first: Enter second: GCD is 1
```

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QUESTIONS

Q1

Find smallest and largest number

10 marks

Q2

Missing Number

10 marks

Q3

Second largest number

10 marks

Q4

Most repeated number

10 marks

Q5

Unique elements

10 marks

Q6

Rotate an array by 'N' positions

10 marks

Q7

Remove all duplicates in an

Q2

Missing Number

10 marks

WAP to find missing number in an array of shuffled order

input: 4 2 1 6 5

output:3

Your Code

```
1 arr=list(map(int,input().split()))
2 n=len(arr)+1
3 total = n*(n+1)//2
4 arr_sum = sum(arr)
5 print(total - arr_sum)
```

Input

4 2 1 6 5

Output

3

Run

Save

Submit

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GCD

10 marks

Q3

sum to a target value

10 marks

Q4

Bitwise Operators

10 marks

Q5

binary search

10 marks

5/5 answered

seen = [] pairs = []

for num in ____:

complement = target - ____

if complement in ____:

pairs.append((complement, num))

____.add(num)

return ____

data = [int(x) for x in input("Enter numbers: ").split()]

t = int(input("Target: "))

result = ____ (data, t)

print("Pairs:", result)

Your Answer

```
set()
lst
num
seen
seen
pairs
```

Save Answer

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Python Programming

QUESTIONS

Q1

Find smallest and largest number

10 marks

Q2

Missing Number

10 marks

Q3

Second largest number

10 marks

Q4

Most repeated number

10 marks

Q5

Unique elements

10 marks

Q6

Rotate an array by 'N' positions

10 marks

Q7

Remove all duplicates in an

Q3

Second largest number

10 marks

WAP to find second largest number in an array

INPUT: 12 5 18 3 25 9

OUTPUT: 18

Your Code

```
1 arr = list(map(int, input().split()))
2 arr = list(set(arr))
3 arr.sort()
4 print(arr[-2])
```

Input

12 15 18 3 25 9

Output

18

Run

Save

Submit

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Python Programming

QUESTIONS

Q1

Harshad number

10 marks

Q2

GCD

10 marks

Q3

sum to a target value

10 marks

Q4

Bitwise Operators

10 marks

Q5

Binary search

10 marks

5/5 answered

Q4

Bitwise Operators

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)

Your Answer

step 1:
a = 0b1101 = 13
b = 0o25 = 21
c = 0*1A = 26
step 2:
a & b = 13 & 21 = 5

Save Answer

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🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def rotated_binary_search(arr, target):
2     low = 0
3     high = len(arr) - 1
4
5     while low <= high:
6         mid = (low + high) // 2
7
8         if arr[mid] == target:
9             return mid
10
11        if arr[low] <= arr[mid]:
12            if target >= arr[low] and target < arr[mid]:
13                high = mid - 1
14            else:
15                low = mid + 1
16        else:
17            if target > arr[mid] and target <= arr[high]:
18                low = mid + 1
19            else:
20                high = mid - 1
21
22        return -1
23
24 data = [4, 5, 6, 7, 0, 1, 2]
25 t = int(input("target: "))
```

Test Input

0

Output

Target: Found at: 4

Pending manual review

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ENGINEERING

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PYTHON PROGRAMMING

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192313059RC

10 marks

Q2
Missing Number
10 marks

Q3
Second largest number
10 marks

Q4
Most repeated number
10 marks

Q5
Unique elements
10 marks

Q6
Rotate an array by 'N' positions
10 marks

Q7
Remove all duplicates in an array
10 marks

Q8
Print all 0's followed by non-zero without changing the order

INPUT: 2 4 6 2 5 3 1 2
OUTPUT: 2

Your Code

```
1 arr = list(map(int, input().split()))
2 count = {}
3 for i in arr:
4     count[i] = count.get(i, 0) + 1
5     max_count = max(count.values())
6     for key in count:
7         if count[key] == max_count:
8             print(key)
9             break
```

Input

2 4 6 2 5 3 1 2

Output

2
2
2
2
2

Run

Save

Submit

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Extensions

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Q2 Missing Number
10 marks

Q3 Second largest number
10 marks

Q4 Most repeated number
10 marks

Q5 Unique elements
10 marks

Q6 Rotate an array by 'N' positions
10 marks

Q7 Remove all duplicates in an array
10 marks

Q8 Print all 0's followed by non-zero without changing the order
10 marks

Your Code

```
1 arr = list(map(int,input().split()))
2 for i in arr:
3     if arr.count(i) == 1:
4         print(i, end=" ")
```

Input

2 4 6 2 5 6 7

Output

4 5 7

Run Save Submit

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Class Work(23.07.26)

Back

QUESTIONS

Q1 Find smallest and largest number
10 marks

Q2 Missing Number
10 marks

Q3 Second largest number
10 marks

Q4 Most repeated number
10 marks

Q5 Unique elements
10 marks

Q6 Rotate an array by 'N' positions
10 marks

Q6 Rotate an array by 'N' positions 10 marks

WAP to rotate an array by 'N' positions
INPUT: 2 5 6 1 3 4
OUTPUT: 6 1 3 4 2 5

Your Code

```
1 arr = list(map(int,input().split()))
2 n = arr[0]
3 arr = arr[1:]
4 n = n % len(arr)
5 result = arr[n:] + arr[:n]
6 print(*result)
```

Input

2 5 6 1 3 4

Output

1 3 4 5 6

Run Save Submit

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Extensions

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Class Work(23.07.26)

← Back

QUESTIONS

Q1

Find smallest and largest number

10 marks

Q2

Missing Number

10 marks

Q3

Second largest number

10 marks

Q4

Most repeated number

10 marks

Q5

Unique elements

10 marks

Q6

Rotate an array by 'N' positions

10 marks

Q8

Print all 0's followed by non-zero without changing the order

10 marks

WAP to print all 0's followed by non-zero without changing the order

INPUT: 1 0 4 0 2 5 0 7 0 6 0

OUTPUT: 0 0 0 0 0 1 4 2 5 7 6

Your Code

```
1 arr = list(map(int, input().split()))
2
3 zero = []
4 nonzero = []
5
6 for i in arr:
7     if i == 0:
8         zero.append(i)
9     else:
10        nonzero.append(i)
11
12 print(" ".join(zero + nonzero))
```

Input

1 0 4 0 2 5 0 7 0 6 0

Output

0 0 0 0 0 1 4 2 5 7 6

Visible Test Cases

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Extensions

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PYTHON PROGRAMMING

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Q2

Missing Number

10 marks

✓

Q3

Second largest number

10 marks

✓

Q4

Most repeated number

10 marks

✓

Q5

Unique elements

10 marks

✓

Q6

Rotate an array by 'N' positions

10 marks

✓

Q7

Remove all duplicates in an array

10 marks

✓

Q8

Print all 0's followed by non-zero without changing the order

10 marks

✓

output: malayalam is a palindrome
input: saveetha
output: saveetha is not a palindrome

Your Code

```
1 s = input()
2
3 if s == s[::-1]:
4     print(s, "is a palindrome")
5 else:
6     print(s, "is not a palindrome")
```

Submitted

Input

malayalam
saveetha

Output

malayalam is a palindrome

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

10.00 / 10 marks

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Extensions

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Verify that it's you

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192313059KC

Q2

Missing Number

10 marks

✓

Q3

Second largest number

10 marks

✓

Q4

Most repeated number

10 marks

✓

Q5

Unique elements

10 marks

✓

Q6

Rotate an array by 'N' positions

10 marks

✓

Q7

Remove all duplicates in an array

10 marks

✓

Q8

Print all 0's followed by non-zero without changing the order

10 marks

✓

output: malayalam is a palindrome
input: saveetha
output: saveetha is not a palindrome

Your Code

```
1 s = input()
2
3 if s == s[::-1]:
4     print(s, "is a palindrome")
5 else:
6     print(s, "is not a palindrome")
```

Submitted

Input

malayalam
saveetha

Output

malayalam is a palindrome

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

10.00 / 10 marks

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10 marks

Q2

Missing Number

10 marks

10 marks

Q3

Second largest number

10 marks

10 marks

Q4

Most repeated number

10 marks

10 marks

Q5

Unique elements

10 marks

10 marks

Q6

Rotate an array by 'N' positions

10 marks

10 marks

Q7

Remove all duplicates in an array

10 marks

10 marks

Q8

Print all 0's followed by non-zero without changing the order

10 marks

INPUT: Hello World 123

OUTPUT:

Vowels:3

Consonants:7

Digits:3

Spaces:2

Your Code

```
1 s = input()
2
3 v = c = d = sp = 0
4
5 for i in s:
6     if i.lower() in "aeiou":
7         v += 1
8     elif i.isalpha():
9         c += 1
10    elif i.isdigit():
11        d += 1
12    elif i == " ":
13        sp += 1
14
15 print("Vowels:", v)
16 print("Consonants:", c)
17 print("Digits:", d)
18 print("Spaces:", sp)
```

Input

Hello World 123

Output

Vowels: 3

Consonants: 7

Digits: 3

Spaces: 2

Visible Test Cases

Test Case 1

0.00 / 10 marks

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10 marks

Q2

Missing Number

10 marks

10 marks

Q3

Second largest number

10 marks

10 marks

Q4

Most repeated number

10 marks

10 marks

Q5

Unique elements

10 marks

10 marks

Q6

Rotate an array by 'N' positions

10 marks

10 marks

Q7

Remove all duplicates in an array

10 marks

10 marks

Q8

Print all 0's followed by non-zero without changing the order

10 marks

output: malayalam is a palindrome

input: saveetha

output: saveetha is not a palindrome

Your Code

```
1 s = input()
2
3 if s == s[::-1]:
4     print(s, "is a palindrome")
5 else:
6     print(s, "is not a palindrome")
```

Input

malayalam

saveetha

Output

malayalam is a palindrome

Visible Test Cases

Test Case 1

Test Case 2

10.00 / 10 marks

Submitted